

Tutorial - 2

A mobile service Provider wants to classify SMS message as spam & Not spam using a simple neural network (Perceptron)

x_1 = Specific words and Phrases

x_2 = Links and Attachments

x_3 = Bias $\rightarrow$ constant

control the boundary of models

x_1 x_2 x_3

target (y) (Yes/No) question

4 2 1

1 0 1

6 3 1

2

$$1 + 2 + 0 = 3$$

$$0$$

$$0 < 5$$

$$1$$

$$\text{if } z < 0 \text{ then } y = 0 \text{ else } y = 1$$

① Inputs : x_1 & x_2

② Weights : w_1 & w_2

③ Bias (b)

④ Weighted sum : $z = x_1 w_1 + x_2 w_2 + b$

$$= \sum_{i=1}^2 x_i w_i + b$$

⑤ Activation function (Here : step function)

$$y^n = \begin{cases} 0; & z < 0 \\ 1; & z \geq 0 \end{cases}$$

⑥ Updated weights.

$$w_i^{\text{New}} = w_i^{\text{Old}} + \eta (y - y^n) x_i$$

η = Learning rate = ①

y = actual target / True target.

y^n = Predicted output / target.

$$x_1 = 4, x_2 = 2, b = 1$$

Let $w_1 = 0, w_2 = 0, b = 1$

Inputs =

$z = ?$

$$z = w_1 x_1 + w_2 x_2 + b$$

$$z = 0(4) + 0(2) + 1$$

$z = 1$ so check $y^n = 1$

the y^n is matched with target (y) in 1st iteration

iteration = epoch $\rightarrow 10$

③ An email Service Provider wants to classify incoming emails as spam or not spam using a single network model.

The training dataset is shown below

$x_1 \quad x_2 \quad x_3 \quad \text{Target (y)}$

5 3 0

1 0 0

2 5 1

2 1 1

6 4 1

① Inputs

$$x_1 = 5, x_2 = 3, b = 1$$

Let $w_1 = 0, w_2 = 0, b = 1$

$$z = w_1 x_1 + w_2 x_2 + b$$

$$z = 0(5) + 0(3) + 1$$

$$z = 1 \text{ so } y^n = 1$$

$$w_{1 \text{ New}} = w_{1 \text{ old}} + \eta(y - y^n) x_1$$

$$w_{1 \text{ New}} = 0 + 1(0 - 1) 5$$

$$w_{1 \text{ New}} = -5$$

$$w_{2 \text{ New}} = w_{2 \text{ old}} + \eta(y - y^n) x_2 = 0 + (0 - 1) 3$$

$$w_{2 \text{ New}} = -3$$

$$w_1 = -5, w_2 = -3, \eta = 1, x_1 = 1, x_2 = 0, b = 1$$

$$z = w_1 x_1 + w_2 x_2 + b$$

$$= (-5)(1) + (-3)(0) + 1$$

$$= -5 + 1$$

$$\boxed{z = -4} \quad \boxed{y^n = 0} \rightarrow \text{the output of old is Predicted}$$

$$w_{\text{New}} = (-5) + 1(y - y^n)x_1$$

$$= (-5) + 1(0 - 0)(1)$$

$$\boxed{w_{\text{New}} = -5}$$

If we update weights after output is correct then the same weights will come.

An e-commerce company Predicts whether a customer will make a Purchase based on time spent on website (minutes) and number of items viewed.

Dataset:

Time on site (x_1)	items viewed (x_2)	Biased (x_3)	Target (Purchase)
2	3		0
15	25		1
8	12		1
3	5		0
12	20		1

① Inputs:

$$x_1 = 2, x_2 = 3, b = 1, w_1 = 0, w_2 = 0, \eta = 1$$

$$z = w_1 x_1 + w_2 x_2 + b$$

$$= (0)(2) + (0)(3) + 1$$

$$z = 1 \Rightarrow \boxed{y^n = 1}$$

$$w_{1, \text{New}} = w_{1, \text{old}} + \eta (y - y^n) x_1$$

$$w_1 = 0 + 1(0-1)2$$

$$w_1 = -2$$

$$w_{2, \text{New}} = w_{2, \text{old}} + \eta (y - y^n) x_2$$

$$w_2 = -3$$

2nd case

$$w_1 = -2, w_2 = -3, b = 1, x_1 = 15, x_2 = 25, \eta = 1$$

$$Z = w_1 x_1 + w_2 x_2 + b$$

$$= (-2)(15) + (-3)(25) + 1$$

$$= -30 + -75 + 1$$

$$= -104 \Rightarrow y^n = 0$$

$$w_{1, \text{New}} = w_{1, \text{old}} + \eta (y - y^n) x_1$$

$$w_1 = -2 + 1(1-0)15$$

$$= -2 + 15$$

$$w_1 = 13$$

$$w_{2, \text{New}} = w_{2, \text{old}} + \eta (y - y^n) x_2$$

$$= -3 + 1(1-0)25$$

$$= -3 + 25$$

$$w_2 = 22$$

$$w_1 = 13, w_2 = 22, \eta = 1$$

$$x_1 = 8, x_2 = 12, b = 1$$

$$Z = w_1 x_1 + w_2 x_2 + b$$

$$Z = (13)(8) + (22)(12) + 1$$

$$Z = 104 + 246 + 1$$

$$Z = 351$$

$$y^n = 1$$

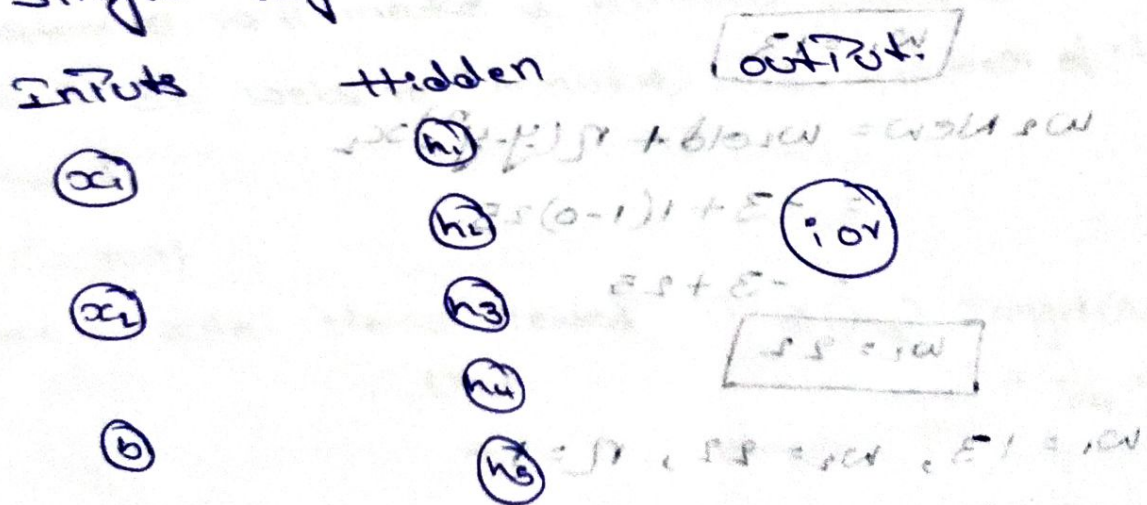
At the third row we have Predicted.

An email service Provider wants to classify incoming emails as spam & Not-spam using a single layer network model with 5 hidden neurons. The training dataset is shown below

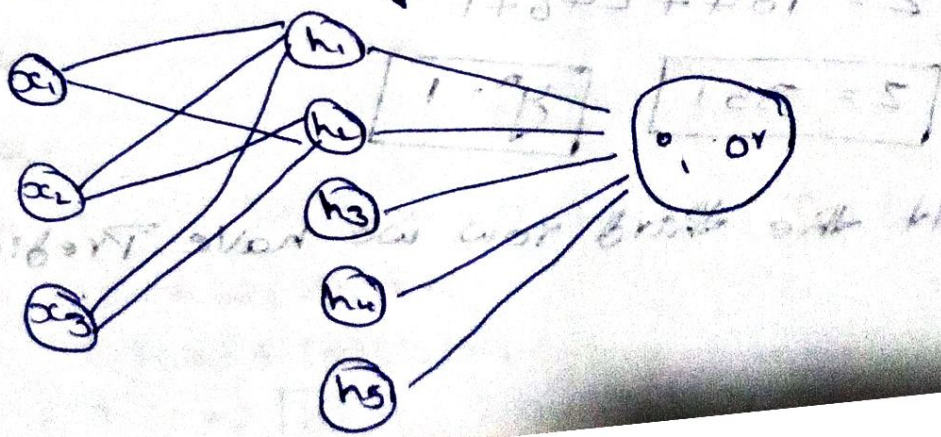
x_1	x_2	x_3	Target (Y)
5	3	1	1
1	0	1	0
2	5	1	1
2	1	1	0
6	4	1	1

Hidden layers are which is located like a mediator b/w inputs and outputs.

single layer neural network.



everything is works/Performed by hidden neuron. each hidden neuron is connected to all the inputs (x_1, x_2, b) in a single input layer.



Ex: $z = w_1 x_1 + w_2 x_2 + b$ $w_1 = 0.1, w_2 = -0.2, b = 1$ $x_1 = 5, x_2 = 3$ $z = 0.5$

into 2 two output nodes z_1 and z_2 $z_1 = 0.5$

h_1 0.1 -0.2 1

h_2 0.3 0.1 1

h_3 0.2 -0.2 1

h_4 0.4 0.1 1

$h_1 = (0.1)(5) + (-0.2)(3) + 1 = 0.5$

$z_1 = w_1 x_1 + w_2 x_2 + b$
 $= (0.1)(5) + (-0.2)(3) + 1 = 0.5$

$z_2 = (0.3)(5) + (0.1)(3) + 1 = 2.8$

$h_2 =$

$z_2 = w_1 x_1 + w_2 x_2 + b = 2.8$

$z = (0.1)(5) + (-0.2)(3) + 1 = 0.5$

$h_3 =$

$z_3 = (0.3)(5) + (0.1)(3) + 1 = 2.8$

$h_4 =$

$z_4 = (0.2)(5) + (-0.2)(3) + 1 = 1.4$

$h_5 =$

$z_5 = (0.4)(5) + (0.1)(3) + 1 = 3.3$

$h_1 = \sigma(z_1) = \frac{1}{1 + e^{-z_1}} = \frac{1}{1 + e^{-0.5}} = 0.74$

$h_2 = \sigma(z_2) = \frac{1}{1 + e^{-z_2}} = \frac{1}{1 + e^{-2.8}} = 0.72$

$h_3 = \sigma(z_3) = \frac{1}{1 + e^{-z_3}} = \frac{1}{1 + e^{-2.8}} = 0.95$

$h_4 = \sigma(z_4) = \frac{1}{1 + e^{-z_4}} = \frac{1}{1 + e^{-1.4}} = 0.80$

$h_5 = \sigma(z_5) = \frac{1}{1 + e^{-z_5}} = \frac{1}{1 + e^{-3.3}} = 0.97$

$$Z_{out} = h_1 v_1 + h_2 v_2 + h_3 v_3 + h_4 v_4 + h_5 v_5$$

$$\sum_{i=1}^N = h_i v_i \quad w \rightarrow \text{hidden layers and inputs}$$

$$v \rightarrow \text{hidden layers and outputs}$$

$$v = 0.1, 0.2, 0.3, 0.4, 0.5, -0.1, -0.5 \text{ any value}$$

$$\sigma(Z_{out}) = ?$$

$$(Z_{out}) = (0.74)(0.1) + (0.72)(0.2) + (0.95)(0.3) + (0.80)(-0.2) + 0.97(0.5)$$

$$\sigma(Z_{out}) = \frac{1}{1 + e^{(-0.828)}} \Rightarrow 0.848$$

$$\sigma(Z_{out}) \Rightarrow 0.999 \approx 1.0$$

$$\sigma(Z_{out}) > 0.5 \quad \boxed{y^n = 1}$$

$$\text{if } Z_{out} \begin{cases} \geq 0.5 \Rightarrow 1 \\ < 0.5 \Rightarrow 0 \end{cases} \text{ Mission will detect.}$$

$$w_i \text{ New} = w_i \text{ Old} + \eta (y - y^n) x_i$$

$$v_i \text{ New} = v_i \text{ Old} + \eta (y - y^n) h_i$$

If the output is less than 0.5 then we want to update the w, v values both/any one of it.

A mobile service Provider wants to classify sms message as spam or Not spam using a single-layer neural network with 5 hidden neurons in the layer.

x_1 = specific words and Phrases.

x_2 = links and attachments.

x_3 = bias

x_1	x_2	x_3	Target (y)
4	2	1	1
1	0	1	0
6	3	1	1
2	1	1	0

Inputs

H.L

Output

x_1

x_2

x_3 (b)

h_1

h_2

h_3

h_4

h_5

0
1

h_1

$$Z_1 = w_1 x_1 + w_2 x_2 + b$$

$$Z = (0.1)(4) + (0.2)(2) + 1 = 1.8$$

h_2

$$Z_2 = w_1 x_1 + w_2 x_2 + b$$

$$= (-0.2)(4) + (0.3)(2) + 1 = 0.8$$

h3

$$z_3 = w_1 x_1 + w_2 x_2 + b.$$

$$= (0.4)(4) + (-0.1)(2) + 1 \Rightarrow 2.4.$$

h4

$$z_4 = w_1 x_1 + w_2 x_2 + b.$$

$$= (0.3)(4) + (0.2)(2) + 1 \Rightarrow 2.6.$$

h5

$$z_5 = w_1 x_1 + w_2 x_2 + b.$$

$$= (0.2)(4) + (-0.4)(2) + 1 \Rightarrow 1.$$

$$h_1 = \sigma(z_1) = \frac{1}{1+e^{-z_1}} = \frac{1}{1+e^{(-1.8)}} = 0.86$$

$$h_2 = \sigma(z_2) = 0.69.$$

$$h_3 = \sigma(z_3) = 0.92.$$

$$h_4 = \sigma(z_4) = 0.93.$$

$$h_5 = \sigma(z_5) = 0.73$$

$$y = 0.1, y_2 = -0.2, y_3 = 0.3, y_4 = 0.4, y_5 = -0.3$$

$$z_{out} = h_1 y_1 + h_2 y_2 + h_3 y_3 + h_4 y_4 + h_5 y_5.$$

$$= 0.86(0.1) + 0.69(-0.2) + (0.92)(0.3) + 0.93(0.4) + 0.73(-0.3)$$

$$= 0.38$$

$$\sigma(z_{out}) = \frac{1}{1+e^{-(0.38)}}$$

$$\sigma(z_{out}) = 0.59 \approx 0.60$$

$$\begin{aligned} \sigma < 0.5 &\rightarrow y^n = 0 \\ \sigma \geq 0.5 &\rightarrow y^n = 1 \end{aligned}$$

$$\Rightarrow \sigma(z_{out}) \geq 0.5$$

$$\text{so } y^n = 1$$

y^n is y Predict

$$V_i \text{ new} = V_i \text{ old} + \eta (y - y^n) h_i$$

$\downarrow$ $\downarrow$ $\rightarrow$
 Learning Rate true Predicted.

$$W_i \text{ new} = W_i \text{ old} + \eta (y - y^n) x_i$$

We can update both V, W (or) any one of it, if output not Predicted.

$$\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

x x x